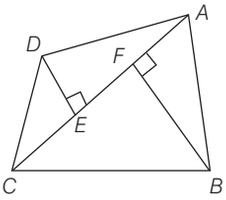
## 12.4 **Quadrilaterals**

A four sided closed figure is called a quadrilateral. It is denoted by symbol ‘ ’.

**Properties**

1. Sum of four interior angles is 360°.
2. The figure formed by joining the mid-points of a quadrilateral is a parallelogram.




3. The sum of opposite sides of a quadrilateral circumscribed about a circle, is always equal.
4. Area of quadrilateral $= \frac{1}{2} \times$ one of the diagonals × sum of the perpendiculars drawn to the diagonals from the opposite vertices. *i.e.,* $A\ (ABCD) = \frac{1}{2} \times AC \times (DE + BF)$

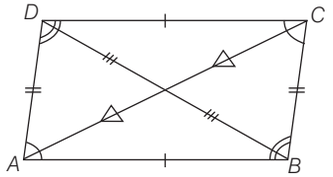
### *Types of Quadrilaterals*

#### *Parallelogram*

A quadrilateral whose opposite sides are parallel.

**Properties**

1. The opposite sides are parallel and equal.




2. Opposite angles are equal.
3. Sum of any two adjacent angles is 180°.
4. Diagonals bisect each other.



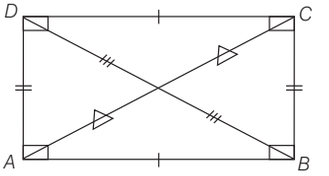

5. Diagonals need not be equal in length.
6. Diagonals need not bisect at right angle.
7. Diagonals need not bisect angles at the vertices.
8. Each diagonal divides a parallelogram into two congruent triangles.
9. Lines joining the mid-points of the adjacent sides of a **quadrilateral** form a **parallelogram.**
10. Lines joining the mid-points of of the adjacent sides of a **parallelogram** is a **parallelogram.**
11. The parallelogram that is inscribed in a circle is a rectangle.
12. The parallelogram that is circumscribed about a circle is a rhombus.
13. (a) Area of a parallelogram = base × height
    (b) Area of parallelogram
    = product of any two adjacent sides × sine of the included angles.
    $= AB \times AD \times \sin\theta$
14. Perimeter of a parallelogram $= 2$ (sum of any two adjacent sides)
15. $(AC)^2 + (BD)^2 = (AB)^2 + (BC)^2 + (CD)^2 + (AD)^2$
    $= 2\,(AB^2 + BC^2)$
16. Parallelogram that lie on the same base and between the same parallel lines are equal in area.
17. Area of a triangle is half of the area of a parallelogram which lie on the same base and between the same parallel lines.
18. A parallelogram is a rectangle if its diagonals are equal.

#### *Rectangle*

A parallelogram in which all the four angles at vertices are right (*i.e.,* 90°), is called a rectangle.

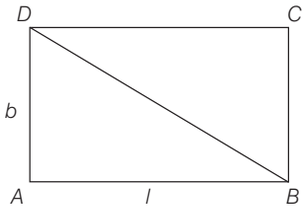
**Properties**

1. Opposite sides are parallel and equal.
2. Opposite angles are equal and of 90°.




3. Diagonals are equal and bisect each other, but not necessarily at right angles.
4. When a rectangle is inscribed in a circle, the diameter of the circle is equal to the diagonal of the rectangle.
5. For the given perimeter of rectangles, a square has maximum area.
6. The figure formed by joining the mid-points of the adjacent sides of a rectangle is a rhombus.

7. The quadrilateral formed by joining the mid-points of intersection of the angle bisectors of a parallelogram is a rectangle.
8. Every rectangle is a parallelogram.
9. Area of a rectangle = length × breadth $(= l \times b)$



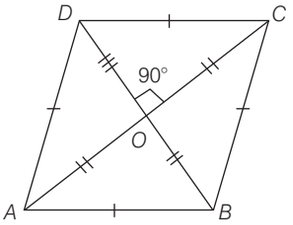

10. Diagonals of a rectangle $= \sqrt{l^2 + b^2}$
11. Perimeter of a rectangle $= 2(l + b)$
    $l \rightarrow$ length and $b \rightarrow$ breadth

## Rhombus

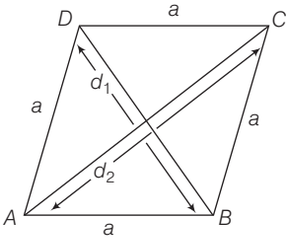
A parallelogram in which all sides are equal, is called a rhombus.

**Properties**

1. Opposite sides are parallel and equal.
2. Opposite angles are equal.
3. Diagonals bisect each other at right angle, but they are not necessarily equal.




4. Diagonals bisect the vertex angles.
5. Sum of any two adjacent angles is 180°.




6. Figure formed by joining the mid-points of the adjacent sides of a rhombus is a rectangle.
7. A parallelogram is a rhombus if its diagonals are perpendicular to each other.
8. (a) Area of a rhombus $= \frac{1}{2} \times$ product of diagonals

$$= \frac{1}{2} \times d_1 \times d_2$$

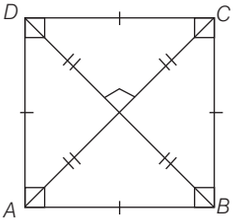
(b) Area of a rhombus = Product of adjacent sides × sine of the included angle.

## Square

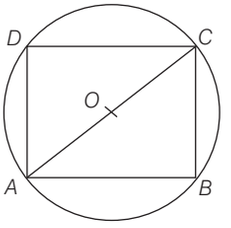
A rectangle whose all sides are equal or a rhombus whose all angles are equal is called a square. Thus each square is a parallelogram, a rectangle and a rhombus.

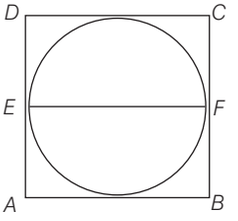
**Properties**

1. All side are equal and parallel.
2. All angles are right angles.
3. Diagonals are equal and bisect each other at right angle.

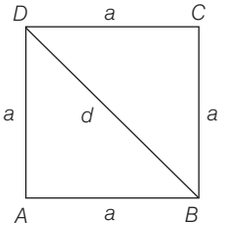



4. Diagonal of an inscribed square is equal to diameter of the inscribing circle.
5. Side of a circumscribed square is equal to the diameter of the inscribed circle.




6. The figure formed by joining the mid-points of the adjacent sides of a square is a square.
7. Area $= (\text{side})^2 = a^2 = \frac{(\text{diagonal})^2}{2} = \frac{d^2}{2}$



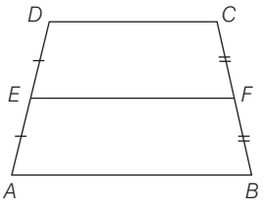

8. Diagonal = side $\sqrt{2} = a\sqrt{2}$
9. Perimeter = 4 × side = $4a$

## Trapezium

A quadrilateral whose only one pair of sides is parallel and other two sides are not parallel.

**Properties**

1. The line joining the mid-points of the oblique (non-parallel) sides is half the sum of the parallel sides and is called the median.



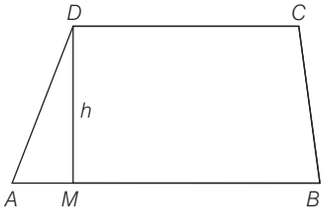

(*i.e.,* Median $= \frac{1}{2} \times$ sum of parallel sides

$= \frac{1}{2} \times (AB + DC) = EF)$

2. If the non-parallel sides are equal then the diagonals will also be equal to each other.
3. Diagonals intersect each other proportionally in the ratio of lengths of parallel sides.
4. By joining the mid-points of adjacent sides of a trapezium four similar triangles are obtained.
5. If a trapezium is inscribed in a circle, then it is an isosceles trapezium with equal oblique sides.
6. Area of trapezium $= \frac{1}{2} \times$ (sum of parallel sides $\times$ height)

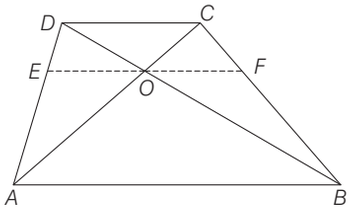
$$= \frac{1}{2} \times (AB + CD) \times h$$

7. $AC^2 + BD^2 = BC^2 + AD^2 + 2AB \cdot CD$




8. If a line segment $EF$, which is parallel to the two parallel sides of a trapezium $ABCD$, passes through a point $O$, which is the point of intersection of the two diagnonals $AC$ and $BD$, where point $E$ lies on $AD$ and point $F$ lies on $BC$, we have

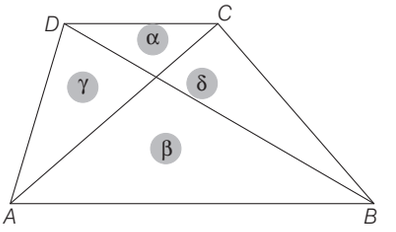
(i) Point $O$ bisects the line $EF$. (ii) $EF = \frac{2(AB)(CD)}{AB + CD}$




9. In a trapezium $ABCD$, as shown in the diagram, where $AB$ is parallel to $CD$ and the diagonals $AC$ and $BD$ intersect each other at $O$, and if the total area of the trapezium is $A$, we have

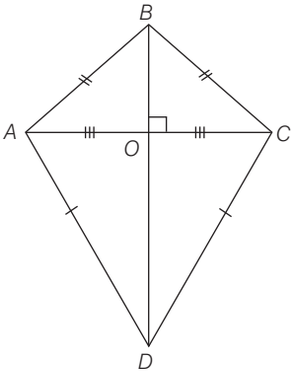
(i) $\sqrt{A} = \sqrt{\alpha} + \sqrt{\beta}$

(ii) $\gamma = \delta = \sqrt{\alpha \cdot \beta}$




## Kite

In a kite two pairs of adjacent sides are equal.



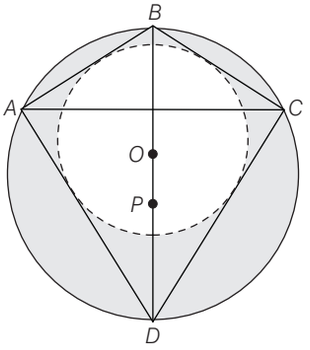

**Properties**

1. $AB = BC$ and $AD = CD$
2. Diagonals, $AC$ and $BD$, are the perpendicular bisectors.
3. Longer diagonal ($BD$) bisects the shorter diagonal ($AC$).
4. Longer diagonal ($BD$) is the angle bisector of the pair of opposite angles.
5. Area of kite $= \frac{1}{2} \times$ product of diagonals

$=$ inradius $\times$ semi-perimeter

6. A kite is symmetrical about the longer diagonal $BD$.
7. The longer diagonal ($BD$) divides the kite into two congruent triangles $ABD$ and $CBD$.
8. The shorter diagonal divides the kite into two isosceles triangles $ABC$ and $ADC$.
9. The kites are exactly the quadrilaterals that are both tangential and orthodiagonal.
10. A kite can always inscribe an incircle and the incentre is the intersection of the angular bisectors of kite.

11. The inradius of the incircle
$$r = \frac{\text{Area of kite}}{\text{semiperimeter of kite}}$$
12. If a kite is equilateral, it must be a rhombus.
13. If a kite is equiangular, it must be equilateral and thus it is a square.
14. The kites that are also cyclic quadrilaterals are exactly the ones formed from two congruent right triangles. That is, for these kites the two equal angles on opposite sides of the axis of symmetry are each 90 degrees. These shapes are called right kites and they are in fact bi-centric quadrilaterals.



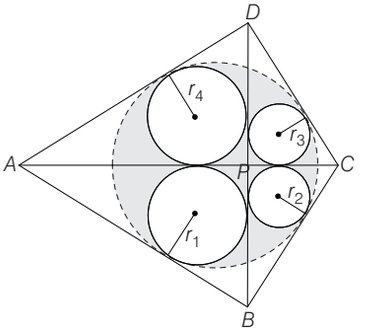

In the above diagram, points $A, B, C, D$ are concyclic, so the quadrilateral $ABCD$ is a concyclic one and the centre of the larger circle is $P$. And the two congruent right triangles are $\Delta BAD$ and $\Delta BCD$. The centre of inscribed circle (smaller one) is $O$. Since the kite has two centers – incentre and circumcentre, that's why a right kite is bi-centric kite.

15. There are only eight polygons that can tile the plane in such a way that reflecting any tile across any one of its edges produces another tile; one of them is a right kite, with 60°, 90°, and 120° angles.

### *Condition for a Tangential Quadrilateral to Become a Kite*

A tangential quadrilateral is a kite if and only if any one of the following conditions is/are true :

(i) The diagonals are perpendicular.
(ii) The area is one half the product of the diagonals.
(iii) The two line segments connecting opposite points of tangency have equal length.
(iv) One pair of opposite tangent lengths has equal length.
(v) The bi-medians have equal length.
(vi) The products of opposite sides are equal.
(vii) The center of the incircle lies on a line of symmetry that is also a diagonal.
(viii) If the diagonals in a tangential quadrilateral $ABCD$ intersect at $P$, and the incircles in triangles $ABP$, $BCP$, $CDP$, $DAP$ have radii $r_1, r_2, r_3$ and $r_4$, respectively, then the quadrilateral is a kite if and only if $r_1 + r_3 = r_2 + r_4$.




(ix) If the excircles to the same four triangles opposite the vertex $P$ have radii $R_1, R_2, R_3$ and $R_4$, respectively, the quadrilateral is a kite if and only if $R_1 + R_3 = R_2 + R_4$.

## **Introductory** *Exercise 12.3*

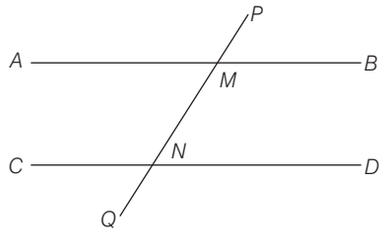
**1.** The measures of the angles of a quadrilateral taken in order are proportional to 1 : 2 : 3 : 4, then the quadrilateral is :
(a) parallelogram (b) trapezium
(c) rectangle (d) rhombus

**2.** Find the measure of largest angle of a quadrilateral if the measures of its interior angles are in the ratio of 3 : 4 : 5 : 6.
(a) 60° (b) 120°
(c) 90° (d) can't be determined

**3.** *ABCD* is a parallelogram, *P* and *Q* are the points on the diagonal *AC* such that *AP* = *QC*, then quadrilateral *BPDQ* is a :
(a) trapezium (b) parallelogram
(c) square (d) none of these

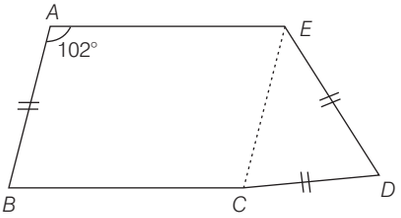
**4.** In a parallelogram *ABCD*, bisectors of consecutive angles *A* and *B* intersect at *P*. Find the measure of $\angle APB$ :
(a) 90° (b) 60°
(c) 120° (d) data insufficient

**5.** $AB$ and $CD$ are two parallel lines and a transversal $PQ$ intersects $AB$ and $CD$ at $M$ and $N$ respectively. The bisector of the interior angles form a quadrilateral :




(a) rectangle (b) square
(c) parallelogram (d) none of these

**6.** In the given figure $AE = BC$ and $AE \parallel BC$ and the three sides $AB$, $CD$ and $ED$ are equal in length. If $m\angle A = 102°$, find measures of $\angle BCD$.




(a) 138° (b) 162°
(c) 88° (d) none of these

**7.** $ABCD$ is a square, $A$ is joined to a point $P$ on $BC$ and $D$ is joined to a point $Q$ on $AB$. If $AP = DQ$ and $AP$ intersects $DQ$ at $R$ then, $\angle DRP$ is :
(a) 60° (b) 120°
(c) 90° (d) can't be determined

**8.** A point $X$ inside a rectangle $PQRS$ is joined to the vertices then, which of the following is true :
(a) $A(\Delta PSX) = A(\Delta RXQ)$
(b) $A(\Delta PSX) + A(\Delta PXQ) = A(\Delta RSX) + A(\Delta RQX)$
(c) $A(\Delta PXS) + A(\Delta RXQ) = A(\Delta SRX) + A(\Delta PXQ)$
(d) None of the above

**9.** $ABCD$ is a parallelogram and m $\angle DAB = 30°$, $BC = 20$ cm and $AB = 20$ cm. Find the area of parallelogram :
(a) 150 $cm^2$ (b) 200 $cm^2$
(c) 400 $cm^2$ (d) 260 $cm^2$

**10.** The length of a side of a rhombus is 10 m and one of its diagonal is 12 m. The length of the other diagonal is :
(a) 15 m (b) 18 m
(c) 16 m (d) can't be determined

**11.** $ABCD$ is a parallelogram and $BD$ is a diagonal. $\angle BAD = 65°$ and $\angle DBC = 45°$,then $m\angle BDC$ is :
(a) 65° (b) 70°
(c) 20° (d) none of these

**12.** If $ABCD$ is a parallelogram in which $P$ and $Q$ are the centroids of $\Delta ABD$ and $\Delta BCD$, then $PQ$ equals :
(a) $AQ$ (b) $AP$ (c) $BP$ (d) $DQ$

**13.** Two parallelograms stand on equal bases and between the same parallels. The ratio of their areas is :
(a) 1 : 1 (b) $\sqrt{2}$ : 1 (c) 1 : 3 (d) 1 : 2

**14.** If a rectangle and a parallelogram are equal in area and have the same base and are situated on the same side, then the ratio of perimeter of rectangle and that of parallelogram is $k$, then $k$ is :
(a) $k > 1$ (b) $k < 1$
(c) $k = 1$ (d) can't be determined

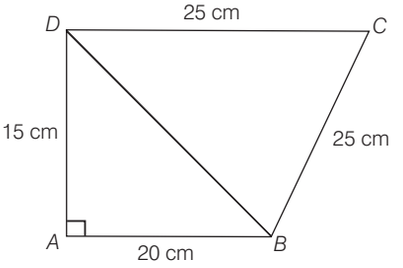
**15.** If area of a parallelogram with sides $l$ and $b$ is $A$ and that of a rectangle with sides $l$ and $b$ is $B$, then :
(a) $A < B$ (b) $A = B$
(c) $A > B$ (d) none of these

**16.** $ABCD$ is a parallelogram and $M$ is the mid-point of $BC$. $AB$ and $DM$ are produced to meet at $N$, then :
(a) $AN = \sqrt{3}AB$ (b) $AN = 2AB$
(c) $AN^2 = \frac{3}{2}AB^2$ (d) $AN^2 = 2AB^2$

**17.** In a rectangle $ABCD$, $P, Q$ are the mid-points of $BC$ and $AD$ respectively and $R$ is any point on $PQ$, then $\Delta ARB$ equals :
(a) $\frac{1}{2}$ ($\square ABCD$) (b) $\frac{1}{3}$ ($\square ABCD$)
(c) $\frac{1}{4}$ ($\square ABCD$) (d) none of these

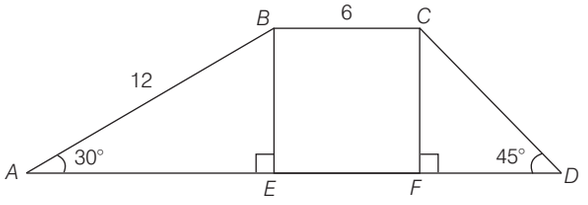
**18.** Diagonals of a parallelogram are 8 m and 6 m respectively. If one of side is 5m, then the area of parallelogram is :
(a) 18 $m^2$ (b) 30 $m^2$
(c) 24 $m^2$ (d) 48 $m^2$

**19.** In the given figure $AD = 15$ cm, $AB = 20$ cm and $BC = CD = 25$ cm. Find the area of $\square ABCD$.



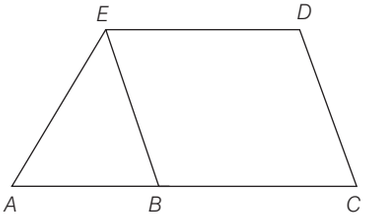

(a) $\frac{25}{4}(24 + 25\sqrt{3})$ $cm^2$
(b) $24(25 + 24\sqrt{3})$ $cm^2$
(c) $\frac{25}{2}(24 + 25\sqrt{3})$ $cm^2$
(d) none of the above

**20.** In the trapezium $ABCD$, $\angle BAE = 30°$, $\angle CDF = 45°$. $BC = 6$ cm and $AB = 12$ cm. Find the area of trapezium.




(a) $18\,(3 + \sqrt{3})\,\text{cm}^2$
(b) $36\sqrt{3}\ \text{cm}^2$
(c) $12\,(3 + 2\sqrt{3})\,\text{cm}^2$
(d) none of the above

**21.** Area of quadrilateral $ACDE$ is 36 $\text{cm}^2$, $B$ is the mid–point of $AC$. Find the area of $\Delta ABE$ if $AC \parallel DE$ and $BE \parallel DC$.




(a) 10 $\text{cm}^2$ (b) 9 $\text{cm}^2$
(c) 12 $\text{cm}^2$ (d) can't be determined

**22.** A square and a rhombus have the same base and the rhombus is inclined at 30°. What is the ratio of area of the square to the area of the rhombus?
(a) $\sqrt{2}:1$ (b) 2 : 1
(c) 1 : 1 (d) $2:\sqrt{3}$

**23.** Find the area of a quadrilateral with sides 17, 25, 30 and 28 cm and one of its diagonal is 26 cm.
(a) 450 $\text{cm}^2$ (b) 360 $\text{cm}^2$
(c) 540 $\text{cm}^2$ (d) 720 $\text{cm}^2$

**24.** $PQRS$ is a parallelogram in which $\angle P = 70°$, $\angle Q = 90°$ and $\angle R = 100°$. How many points in the plane of the quadrilateral are there such that a point is equidistant from its vertices ?
(a) 0 (b) 1
(c) 2 (d) 3

**25.** $ABCD$ is a parallelogram. The diagonals $AC$ and $BD$ intersect at a point $O$. If $E$, $F$, $G$ and $H$ are the mid-points of $AO$, $DO$, $CO$ and $BO$ respectively, then the ratio of $(EF + FG + GH + HE)$ to $(AD + DC + CB + BA)$ is :
(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 1 : 4

**26.** If $ABCD$ is a rhombus, then :
(a) $AC^2 + BD^2 = 4AB^2$
(b) $AC^2 + BD^2 = AB^2$
(c) $AC^2 + BD^2 = 2AB^2$
(d) $2\,(AC^2 + BD^2) = 3AB^2$

**27.** If $P$ is a point within a rectangle $ABCD$, then :
(a) $AP^2 + PC^2 = BP^2 + PD^2$
(b) $AP^2 + BP^2 = PC^2 + PD^2$
(c) $AP + PC = BP + PD$
(d) $AP \times PC = BP \times PD$

**28.** $\square ABCD$ is a parallelogram, $AB = 14$ cm, $BC = 18$ cm and $AC = 16$ cm. Find the length of the other diagonal.
(a) 24 cm (b) 28 cm
(c) 36 cm (d) 32 cm

**29.** In a trapezium $ABCD$, where $AB \parallel CD$ and diagonal $AC$ intersects the other diagonal $BC$ at $O$. Which of the followings are always correct?

(i) $\dfrac{DO}{AO} = \dfrac{CO}{BO}$ (ii) $\dfrac{DO}{OC} = \dfrac{AO}{OB}$

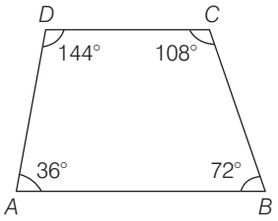
(iii) $\dfrac{DO}{OB} = \dfrac{CO}{OA}$

(a) only (i) and (ii)
(b) only (ii) and (iii)
(c) only (i) and (iii)
(d) only (iii)

**30.** In a trapezium $ABCD$, where $AB \parallel CD$ and diagonals $AC$ intersects the other diagonal $BC$ at $O$. If a line segment $EF$ parallel to $AB$ and $CD$ passes through the point $O$, where $E$ lies on $AD$ and $F$ lies on $BC$, find the length of $EF$, provided $AB = 6$ cm and $CD = 2$ cm.
(a) 3 cm (b) 4 cm
(c) 4.5 cm (d) none of these

## ☑ Introductory Exercise 12.3

**1** $x + 2x + 3x + 4x = 360 \Rightarrow x = 36$




$\therefore$ The angles of quadrilateral (in order) are

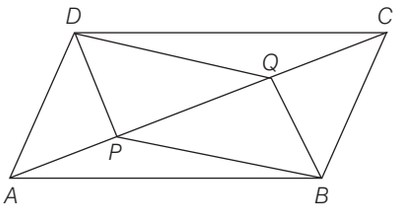
$36°, 72°, 108°, 144°$

Since opposite angles are supplementary, therefore $AB \| CD$. Hence it is a trapezium.

**2** $3x + 4x + 5x + 6x = 360° \Rightarrow x = 20 \quad \therefore \quad 6x = 120°$

Hence, the largest angle of a quadrilateral is 120°.

**3**

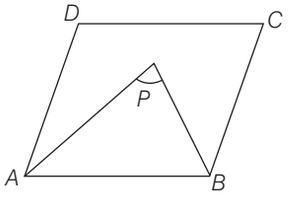



**4** $\angle DAB = 2x \Rightarrow \angle ABC = 2y$

$2x + 2y = 180 \Rightarrow x + y = 90°$

*i.e.,* $\angle PAB + \angle PBA = 90°$

$\therefore \quad \angle APB = 90°$




**5** Prove that opposite sides are parallel, and diagonals are equal. Also all the interior angles are right angle.

**6** $\angle BCE = 102°$ and $\angle ECD = 60°$

$\therefore \angle BCD = 102° + 60° = 162°$

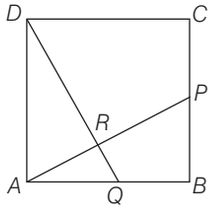
**7** $\because \quad \Delta ADQ \cong \Delta BAP$

and $\angle DCP = 90°$

$\angle CDR$ and $\angle CPR$ are supplementary

$\therefore \quad \angle DCP + \angle DRP = 180°$

$\therefore \quad \angle DRP = 90°$

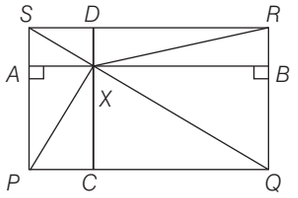



**8** Area of $\Delta SXP = \dfrac{(SP)\times(AX)}{2}$

$$A(\Delta RXQ) = \frac{(RQ)\times(BX)}{2}$$

$$A(\Delta SXR) = \frac{(SR)\times(DX)}{2}$$

$$A(\Delta PXQ) = \frac{(PQ)\times(CX)}{2}$$




$$\text{Area of } \Delta SXP + RXQ = \frac{SP\times AX}{2} + \frac{RQ\times BX}{2}$$

$$= \frac{1}{2}(l\times b) \qquad \ldots(\text{i})$$

where $SP = RQ = b$ and $PQ = SR = l$

and $AX + BX = AB = PQ = SR = l$

$$\text{and Area of } \Delta SXR + PXQ = \frac{(SR)\times(DX)}{2} + \frac{(PQ\times CX)}{2}$$

$$= \frac{(SR)\times(DX)}{2} + \frac{(SR)\times(CX)}{2}$$

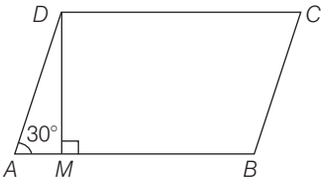
$$= \frac{SR}{2}(DX + CX)$$

$$= \frac{l}{2}\times b = \frac{lb}{2} \qquad \ldots(\text{ii})$$

From eqs. (i) and (ii), we find that option (c) is correct.

**9** $AD = BC = 20$ cm

$AB = 20$




In $\Delta ADM$, $\dfrac{DM}{AD} = \sin 30° = \dfrac{1}{2}$

$\therefore \quad DM = \dfrac{1}{2}AD = 10$ cm

$\therefore$ Area of parallelogram = base × height

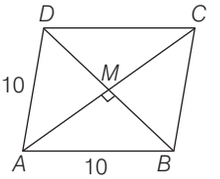
$= 20 \times 10 = 200 \text{ cm}^2$

**10** Let $DB = 12$ m

$\therefore \quad BM = 6$ m

$\therefore \quad AM = \sqrt{10^2 - 6^2}$

$AM = 8$ m



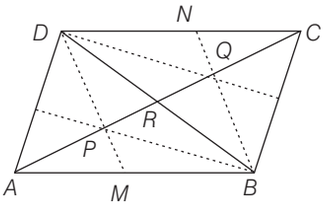

(**Remember:** Diagonals bisects each other at right angles)

(**Bisect:** Divide equally into two parts)

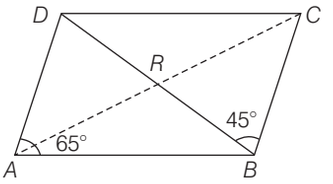
$\therefore \quad AC = 8 \times 2 = 16$ m

**11** $\angle BCD = \angle DAB = 65°$

$\therefore$ $CDB = 180° - (65° + 45°) = 70°$




**12** Let $AC$ and $BD$ bisects each other at $R$, then $AP = 2PR$ and $QC = 2RQ$



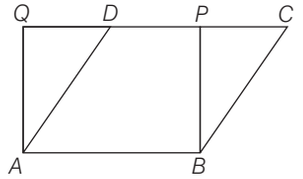

($\because$ $AR$ and $CR$ are the medians of $\Delta ADB$ and $\Delta BDC$)

and $AP = QC$ and $PR = RQ$ ($\because \Delta DAB \cong \Delta BCD$)

$\therefore$ $AP = PQ$

**13** Since their base and perpendicular heights are same.

**14** $ABCD$ is a parallelogram and $ABPQ$ is a rectangle having same base and equal perpendicular height. Therefore the area of both is same.




Now, since $AD > AQ$

and $BC > BP$

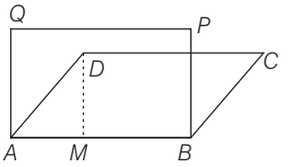
$\therefore$ $2(AB + BC) > 2(AB + BP)$

$\therefore$ Perimeter of parallelogram > perimeter of rectangle.

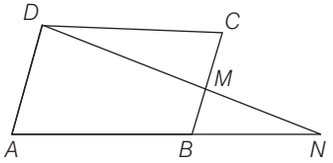
**15** $AB = CD = PQ = l$

and $AQ = BP = AD = CB = b$

Bases of rectangle $ABPQ$ and parallelogram $ABCD$ is same, but perpendicular heights of parallelogram is less than that of rectangle. Therefore area of parallelogram is less than that of rectangle.




**16** $AD = 2BM$ and $AD \parallel BM$

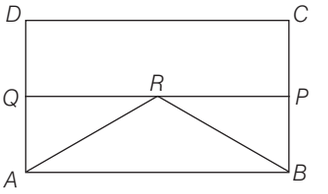



$\therefore$ From mid-point theorem

$\therefore$ $\dfrac{NM}{ND} = \dfrac{NB}{NA} = \dfrac{BM}{AD} = \dfrac{1}{2}$

$\therefore$ $AN = 2AB$

**17** $A(\square ABPQ) = \dfrac{1}{2} A(\square ABCD)$

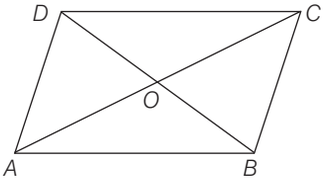



and $A(\Delta ARB) = \dfrac{1}{2} A(\square ABPQ)$

$\therefore$ $A(\Delta ARB) = \dfrac{1}{4} A(\square ABCD)\left[A(\Delta ARB) = \dfrac{1}{2} \times AB \times BP\right]$

**18** Let $BD = 6$ m and $AC = 8$ m

$\therefore$ $AO = 4$ m and $BO = 3$ m




Let $AB = 5$ m $\quad \therefore \angle AOB = 90°$

$\Rightarrow$ $\angle BOC = \angle AOD = \angle DOC = 90°$

$\therefore$ $BC$ is also 5 m.

Hence, $ABCD$ is a rhombus

$\therefore$ Area of rhombus $ABCD = \dfrac{AC \times BD}{2} = \dfrac{6 \times 8}{2} = 24\ \text{m}^2$

**19** $BD = \sqrt{15^2 + 20^2} = 25$ cm

($\Delta ABD$ is right angled triangle)

$\therefore \Delta BCD$ is equilateral triangle.

$\therefore$ Area of $ABCD$ = Area of $\Delta ABD$ + Area of $\Delta BCD$

$= \dfrac{1}{2} \times 15 \times 20 + \dfrac{\sqrt{3}}{4} \times (25)^2 = \dfrac{25}{4}(24 + 25\sqrt{3})\ \text{cm}^2$

**20.** $\dfrac{BE}{AB} = \sin 30° = \dfrac{1}{2}$

$\therefore$ $BE = \dfrac{1}{2} \times AB = 6\ \text{cm} = CF$

and $\dfrac{CF}{DF} = \tan 45° = 1$

$\therefore$ $DF = CF = 6$ cm

and $\dfrac{AE}{AB} = \cos 30° = \dfrac{\sqrt{3}}{2}$

$\therefore$ $AE = 6\sqrt{3}$ cm

$\therefore$ $AD = AE + EF + FD = 6\sqrt{3} + 6 + 6$

$= 6(2 + \sqrt{3})\ \text{cm}^2$

Area of trapezium $ADCB = \dfrac{1}{2} \times (AD + BC) \times BE$

$= \dfrac{1}{2} \times [6(2 + \sqrt{3}) + 6] \times 6$

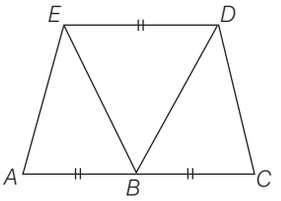
$= \dfrac{6}{2}(2 + \sqrt{3} + 1) \times 6 = 18 \times (3 + \sqrt{3}) = 18(3 + \sqrt{3})\ \text{cm}^2$

**21** $A(\Delta ABE) = A(\Delta BDE)$

$= A(\Delta BCE)$

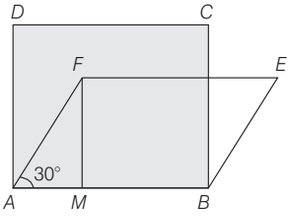
$A(\Delta ABC) = \frac{1}{3} A(\square ACDE)$

$= \frac{1}{3} \times 36 = 12\,\text{cm}^2$




**22** $ABCD$ is a square and $ABEF$ is a rhombus.

$$\frac{FM}{AF} = \sin 30° = \frac{1}{2} \Rightarrow FM = \frac{AF}{2}, \quad AF = AB$$




Area of square $= a^2$ $\quad (AB = AD = a)$

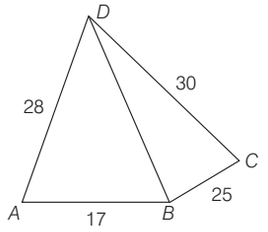
Area of rhombus $= \frac{a \times a}{2}$ $\quad \left(FM = \frac{a}{2}\right)$

(Area of rhombus = base × height)

$\therefore$ $\frac{\text{Area of square}}{\text{Area of rhombus}} = \frac{2}{1}$

**23** Area of quadrilateral = Area of $\Delta ABD$ + Area of $\Delta BCD$

where $BD = 26$ cm




**Hint** Apply Hero's formula :

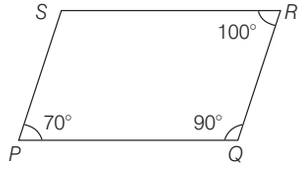
$$\text{Area of } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \frac{a+b+c}{2}$

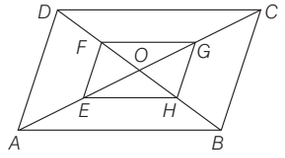
**24** It is not possible.

$\because \angle P + \angle R = 170° \neq 180°$

$\therefore$ The given quadrilateral is not cyclic so there is no any point in the given plane which is equidistant from its vertices.




**25** By mid-point theorem,

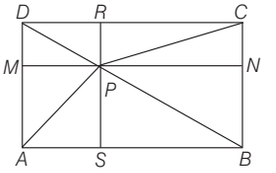



$$\frac{EF}{AD} = \frac{FG}{DC} = \frac{GH}{BC} = \frac{EH}{AB} = \frac{1}{2}$$

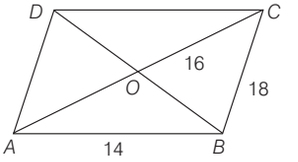
**26** Since diagonals bisect each other at right angles and all sides are equal.

Apply Pythagoras theorem or Apollonius theorem.

**27** Use Pythagoras theorem




**28** Apply Apollonius theorem.



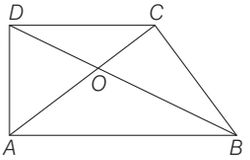

$$AB^2 + BC^2 = 2(BO^2 + AO^2)$$

$\Rightarrow$ $AB^2 + BC^2 = 2(BO^2 + 64)$

$\Rightarrow$ $196 + 324 = 2(BO^2 + 64)$

$\therefore$ $BO^2 = 196 \Rightarrow BO = 14\,\text{cm} \Rightarrow BD = 2BO = 28\,\text{cm}$

**29** Consider the following diagram, where $AB$ is parallel to $CD$ and $AC$ and $BD$ act as transversal lines.




$\angle AOB = \angle COD$ (Opposite angles)

$\angle OAB = \angle OCD$ (Alternate angles)

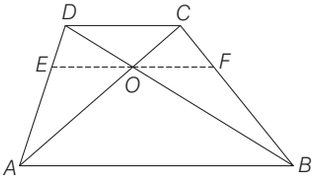
$\angle ABO = \angle CDO$ (Alternate angles)

$\therefore$ $\Delta AOB \sim \Delta COD$ (AAA similarity)

Therefore $\frac{AB}{CD} = \frac{AO}{OC} = \frac{BO}{OD}$

Hence choice (d) is correct.

**30** Let us consider the following diagram. Where $AB \,||\, CD \,||\, EF$. Further, let us consider $AB = m$ and $CD = n$, we have, $\frac{OA}{OC} = \frac{m}{n}$ and $\frac{OB}{OD} = \frac{m}{n}$.




But, $EF = \frac{(AB)n + (CD)m}{m+n} = \frac{(m)n + (n)m}{(m+n)}$

$= \frac{2mn}{m+n} = \frac{2(AB)(CD)}{(AB+CD)}$

Therefore, $EF = \frac{2(AB)(CD)}{(AB+CD)}$

$= \frac{2(6 \times 2)}{6+2} = 3\,\text{cm}$

Hence choice (a) is the correct one.